Deadline Aware Task Submission and Dynamic Virtual Machine Creation Technique in Cloud Computing

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ABSTRACT

The growth of cloud computing has resulted in uneconomic energy consumption, which has negatively impacted the environment through the generation of carbon emissions. One of the most important issues for green cloud environments concerns where to place new virtual machine (VM) requests across physical servers in a way that ensures reduced energy consumption. This paper proposes a deadline based dynamic virtual machine (DBDV) migration algorithm to reduce energy consumption in Cloud datacenters. Virtual machines are classified in three categories: compute intensive task, data intensive task and mix load task. Tasks have been allocated based on similar category of virtual machine. Compute intensive task is assigned to compute intensive virtual machine, data intensive task is assigned to data intensive virtual machine and mix load task is assigned to mix load virtual machine. All tasks are assigned to similar type of virtual machine without classifying virtual machine. The proposed algorithm for virtual machine mapping is based on deadline of task and migration of virtual machine. The proposed algorithm is compared with threshold-based method, PSO and Ant Colony Optimization algorithm and analyzed the result for the energy utilization and number of virtual machine migration as well as completion time. The experimental results show that a DBDV migration algorithm outperforms the existing algorithms in terms of these parameters.

CCS CONCEPTS

• **Network** → Network Services; Cloud Computing.

KEYWORDS

Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), VM migration, Cloud computing, Energy efficiency

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1 INTRODUCTION

The cloud computing model leverages a vast number of virtualized computing resources to provide them as utilities to customers on a pay-as-you-go basis. The growth of cloud computing has resulted in uneconomic energy consumption, which is harmful to the environment due to the huge carbon footprints of cloud datacenters. A typical datacenter consumes as much energy as 25,000 households [1]. An average datacenter can also produce more than 150 million metric tons of carbon annually. Carbon emissions generated through cloud computing worldwide in 2020 are expected to total approximately 670 million metric tons [2]. Additionally, massive volumes of energy consumed at datacenters increase operational costs because when volume increase it requires large number of resources and finally it it requires more electricity and staff to maintain it functional.. In 2013, US datacenters used an estimated 91 billion kilowatt-hours of electricity [2]. Therefore, green cloud computing must be used to protect the environment. Green cloud computing aims to save the environment from datacenters' carbon emissions by reducing energy consumption levels.

One of the most important issues for green cloud environments concerns where to place new virtual machine (VM) requests across physical servers in a way that ensures reduced energy consumption. Many research initiatives have attempted to answer this question in reference to hardware [8] [9] [10] [11], network [12] [13] [14] [15] [17] or software layers [4] [5] [6] [7] [16] [18]. Methods focusing on the latter have been the most commonly used owing to their flexibility and cost effectiveness relative to other approaches. However, most software approaches suffer from single point of failure and poor scalability issues owing to their centralized control schemes [3]. To tackle such problems, bio-inspired heuristics are emerging, are offering more decentralized control and are lessening the probability of failures while increasing scalability levels to support a larger number of nodes in green clouds. Based on the generalization of the algorithm, heuristics approaches can be classified into two categories. The first, problem-independent heuristics approaches also known as metaheuristics (e.g., the Ant Colony Optimization algorithm (ACO)), concern algorithms designed to address a wide range of optimization problems. A meta-heuristic starts with a simple or random solution and then iteratively attempts to improve a candidate solution with regards to a given measure of quality, i.e., objective function. Metaheuristics do not have to deeply adopt to a problem domain, causing them to be generalized and easy to design and to implement solutions. However, the main drawbacks of metaheuristics concern their steep exponential running periods and high levels of power consumption, rendering them advisable to use only when no problem-dependent heuristic is available.

The second approach involves the use of problem-dependent heuristics (PDHs) such as those proposed here. These algorithms are designed to address a specific problem. A PDH is a tailored algorithm that addresses a specific problem while taking its particularities into consideration. It strives to generate a "good-enough" solution one a first attempt, eliminating the need to iteratively optimize it, saving considerable time and processing power but rendering the tool difficult to design and implement [3], [19]. To this end, this paper proposes a problem-dependent deadline based dynamic virtual machine (DBDV) migration algorithm to reduce energy consumption in Cloud datacenters. The algorithm utilizes a distributed scheduling policy to enable each running server to serve the largest possible number of VMs. When the number of tasks running on a virtual machine falls below a certain threshold, corresponding VMs are migrated to another server; then, idle servers can be switched off.

A strictly controlled evaluation framework for the proposed DBDV algorithm was designed with a focus on two experimental variables: the datacenter load and datacenter scale. Our evaluation was based on two performance measures: total datacenter energy consumption, number of VM migration. DBDV performance was compared to that of three well-established green cloud computing benchmarks: Threshold based Method [21], Particle Swarm Optimization [22] and the static Ant Colony Optimization (ACO) [40]. Our results show that DBDV considerably reduces energy consumption levels without compromising other performance measures. Most importantly, DBDV can effectively tolerate heavy workloads that benchmarks are unable to sustain. In summary, the main contributions of this paper are as follows:

- The paper introduces DBDV, a green cloud scheduling algorithm that to the best of our knowledge is the first PDH approach to reference locusts' phase change behaviors to address power consumption problems.
- Following a strictly controlled experimental framework, we evaluate DBDV algorithm's performance based on real workloads and well-established benchmark algorithms.

This paper is organized as follows. Related works are described in Section 2, Section 3 presents the system model developed, and the DBDV algorithm is introduced in Section 4. Section 5 presents the evaluation framework used and discusses our simulation results. Finally, Section 6 concludes the paper and highlights avenues for future research.

2 RELATED WORKS

One of the most important issues related to green cloud environments concerns where to place new VM requests across physical servers in such a way that ensures reduced energy consumption. Many research initiatives have emerged to address this question. However, such efforts remain in their early stages and can be classified into three types: hardware optimization, network optimization and software optimization initiatives. A detailed survey of these approaches is presented in Ismail and Kurdi paper [3].

Hardware optimization techniques reduce energy consumption levels by utilizing flexible hardware that regulates server-computing capabilities by controlling server running frequencies and voltages [24], [25]. Following from this approach, the DVFS technique is presented in [21]. DVFS utilizes a number of special processors that can operate at different voltage and frequency levels. DVFS selects appropriate supply voltages and frequencies of processing elements to minimize energy consumption levels without violating SLA based on VM workload requirements.

Following a similar approach, Cao and Zhu [26] used DVFS servers to determine the optimal frequency level appropriate for each task in a scientific workflow by proposing the Energy-Aware Resource Efficient workflow Scheduling under Deadline constraints (EARES-D). The optimal frequency for executing each task is determined by scaling down the processor frequency under the deadline constraint. VM reuse and idle time shrinking policies are used to improve resource utilization efficiency. However, in general, hardware optimization techniques for green clouds are costly and suffer from poor levels of scalability, as they involve the use of special hardware. Moreover, DVFS implementations suffer from cubic time complexities ($O(n^3)$) as indicated by [27]. More information on hardware optimization techniques is available in [24]. Network optimization techniques are designed to reduce energy consumption levels by minimizing network traffic between servers [2]. An example of this approach is presented in [28], in which two techniques for flow and VM migration are proposed. The first technique generates different disjoint-spanning trees to avoid overlapping paths and selects the least utilized reroute flow path. The second technique migrates VMs from under-/over-utilized servers to the nearest machine based on the network distance. Both techniques have been evaluated via simulations using the Network Simulator NS2 [29] and CloudSim simulator [42]. The results show an improvement in throughput levels but also increased levels of energy consumption relative to the benchmark (i.e., the Shortest Path Bridge (SPB)).

In [30], the Datacenter Energy efficient Network-aware Scheduling (DENS) approach is proposed. Other studies have also focused on reducing energy consumption levels by minimizing network traffic between servers [2], [28], [31], [32]. However, the use of network optimization techniques involves determining network topologies, as they are applicable to specific topologies only, limiting their scalability and flexibility. A full review of network optimization techniques are presented in [33]. The increasing number of users requires resource provisioning strategy. Virtual machine (VM) placement is one of the keys to assure cloud consolidation and reduce power wastage [22]. The authors proposed a solution based on particle swarm optimization (PSO) for the virtual machine placement in the heterogeneous data center. In this work, authors worked to improve the PSO method to fit with virtual machine placement in data centers. Results showed that improved PSO can find the optimal solution for low energy consumption comparing to First-fit, Best-fit, and worst-fit. However, the algorithms include large computation overheads associated with VM allocation or migration and with rescheduling VMs in the optimization phase.

An algorithm for VM placement and migration that addresses both over- and under-utilized servers is proposed in [23]. The static Threshold based on a Minimum Utilization policy algorithm (ThrMu) allocates new VM requests to the most power-efficient server generating the lowest increase in energy consumption, and it then reallocates previously scheduled VMs to lightly loaded and over-utilized servers. The most power-efficient server is selected based on complex computations for power predictions that consume time and resources. The main drawbacks of this approach concern its centralized policy and expensive computations required to predict the power consumption levels of each server prior to VM allocation. Energy efficiency can also be achieved by VM multiplexing so that multiple VMS can be consolidated and provisioned together when their peaks and troughs are temporally unaligned as suggested by Meng *et al.* [34]. In a similar manner, Simao and Veiga proposed a cost model that considers each user's partial utility specification when a CSP needs to transfer resources between VMs. Corresponding results show that this strategy overcomes shortcomings of the classic allocation strategy, which cannot allocate above a certain number of VMs, although the proposed approach consumes more computational power when resources are scarce [20] [35]. A focused survey on software optimization algorithms is presented in [36].

Bio-inspired algorithms for green clouds are increasingly attracting attention owing to their capacities to find "good"' solutions in a cost-effective manner. An energy aware multi-agent firefly inspired algorithm is proposed by Kansal and Chana [37]. It utilizes the VM migration technique to reduce energy consumption by imitating firefly behaviors in performing local and global searches for the best VM and server pair. The proposed approach was evaluated using the CloudSim toolkit. The results show an enhancement in average energy consumption. However, this is reflected negatively in response times [38]. A recent study [39] proposed a cuckoo-inspired energy-aware load balancing technique. The proposed algorithm allocates resources randomly at first. Thereafter, the algorithm calculates a fitness value for each allocated server, and VMs are allocated to a server based on this fitness value. The proposed approach was applied using the CloudSim toolkit, and the results show an improvement in energy consumption outcomes. However, the mapping process of the proposed approach is time consuming. Virtually all of the above algorithms apply centralized control and suffer from single point of failure issues and poor levels of scalability. The decentralization of control typically lessens the probability of failure while increasing scalability levels for a larger number of nodes. A decentralized ant-inspired distributed algorithm is proposed in [40] based on the behaviors of real ants. The algorithm uses the Ant Colony Optimization (ACO) meta-heuristic to determine VM placement. Its goal is to minimize the number of active servers used by maximizing resource utilization to reduce energy consumption. On receiving a VM request, each server (ant) suggests a schedule for fitting the VM. After receiving schedules from all servers, the solution with the lower resource utilization value is selected. The authors developed their own java-based simulation toolkit [40] to evaluate their approach. Their experimental results show energy consumption improvements of up to nearly 4%, which comes at a high computational cost, as stated by the authors. A common problem related to all ACO techniques concerns the fact that they take a long time to converge [41]. Additionally, having each server suggest and send a schedule for each VM request considerably wastes system and network resources. In this paper, we consider a decentralized software optimization approach to ensuring robustness, scalability, and cost effectiveness when designing the DBDV scheduling Algorithm to reduce energy consumed in Cloud datacenters. The proposed DBDV algorithm differs from other green cloud algorithms in the following ways:

- VM scheduling is distributed across servers rather than centralizing scheduling decisions.
- The algorithm exploits servers heterogeneously to reduce power consumption levels.
- The algorithm exploits server capacities to reduce the number of active servers used by migrating VMs from the least loaded server (LLS) to a heavy loaded powerful server to reduce power consumption levels.
- VM allocation and consolidation are performed based on accurate and updated data; each server is responsible for selecting VMs based on a clear account of its load and currently available resources.

The algorithm does not initiate the migration phase until total current free resources available in the datacenter are checked to prevent unprofitable VM migrations.

3 SYSTEM ARCHITECTURE AND PROPOSED ALGORITHM

The datacenter has four components (a VM request queue, a system registry, a server, and as set of VM) as shown in Figure 1

VM request queue: This component receives newly arriving VM requests from the end user. The VM request queue is globally accessed by all servers in the datacenter. VM requests are retrieved in a First in First out (FIFO) manner.

Centralized System Registry (CSR): This component includes information on all system servers, including information their memory capacities, processor types, bandwidth levels, resource utilization patterns, and statuses. The CSR is responsible for calculating and examining the Global Migration Rule and for sending corresponding results to a server when the server requests it. CSR is also responsible for sending LLS information to other servers. It also calculates the consolidation threshold (CT) and broadcasts it to servers during initialization and when the CT changes by a certain percentage.

Servers: This component is the core component of the system; it represents physical servers. Servers consume the most datacenter power, and processors are major energy consumers in each server.

VM set: Three different categories of VMs have been considered.

- Compute Intensive VM: A virtual machine having high MIPS capacity and low bandwidth requirement is classified in this category.
- Data Intensive VM: A virtual machine having high bandwidth requirement and high MIPS capacity is classified as data intensive VM.
- Balanced VM: A virtual machine having low bandwidth requirement and low MIPS capacity is classified as balanced VM.

We have divided the work in mainly two parts. First part is task submission on VM, which ensure task completion within deadline and second part, is VM creation and migration on host to optimize the resource utilization. There are mainly three types of task data intensive, compute intensive and balanced task. Balanced task is a task that is not in compute intensive and not in data intensive.

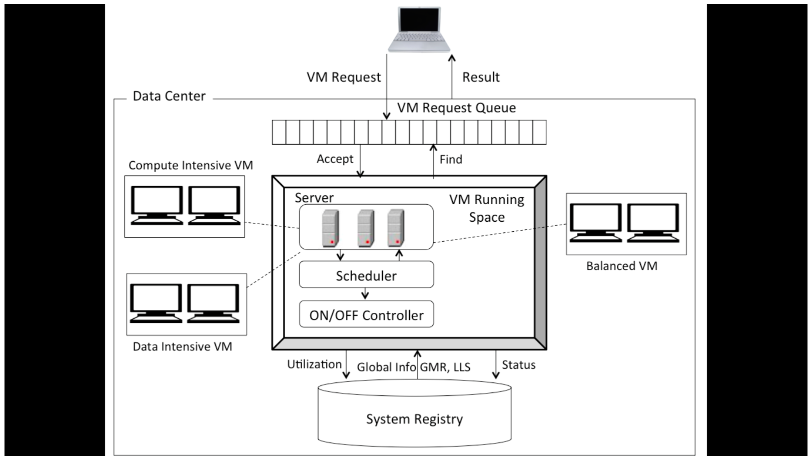

Figure 1: DBDV System Architecture

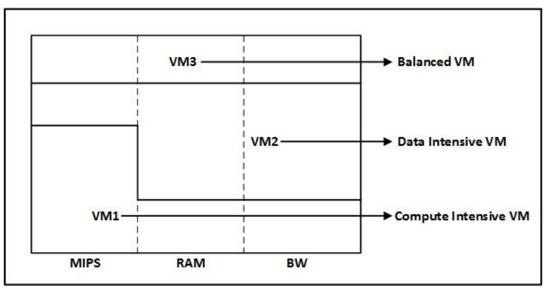

Figure 2: VM Creation on Host

Compute intensive task: A task having high MIPS requirement and low data communication requirement is classified in this category. A task must satisfy the below condition:

$$Task_{MIPS} > TH_{Task\_MIPS} \&\& Task_{RAM} < TH_{Task\_RAM} \quad (1)$$

In equation 1), $\mathrm{Task_{RAM}}$ is the actual RAM requirement by a task and $\mathrm{TH_{Task\_RAM}}$ is threshold value of the RAM. In a similar way, $\mathrm{Task_{MIPS}}$ is the actual MIPS requirement by a task and $\mathrm{TH_{Task\_MIPS}}$ is the threshold number of MIPS for a task.

Data intensive task: A task having high data communication requirement and high MIPS requirement is classified as data intensive task. A task must satisfy the condition:

$$Task_{RAM} > TH_{Task\_RAM} \&\& Task_{MIPS} > TH_{Task\_MIPS} \quad (2)$$

Balanced task: A task having low data communication requirement and low MIPS requirement is classified as balanced task. A task must satisfy the condition:

$$Task_{MIPS} < TH_{Task\_MIPS} \&\& Task_{RAM} < TH_{Task\_RAM} \quad (3)$$

Similarly, VMs also categorized in three parts high MIPS, high RAM and high Bandwidth and balanced VM as shown in figure2. In a similar way, VMs are also classified in three categories in compute intensive VM, Data intensive VM, and balanced VM based on static threshold. Threshold for MIPS, RAM and Band width are $\mathrm{TH_{VM\_MIPS}}$, $\mathrm{TH_{VM\_RAM}}$ and $\mathrm{TH_{VM\_BW}}$.

A. *Task submission*

When user submit task, it will be submitted to broker. Broker will find suitable VM on host to execute the task. Every time broker gets many task requests to execute on cloud from different users. Broker has a responsibility to complete task within the requested deadline of user. Task submission algorithm allocates VM to each task so that maximum task can complete its execution within deadline. Task is submitted to VM if it satisfies following conditions:

$$Deadline > Task_{MIPS}/VM_{Available\_MIPS} \quad (4)$$

In equation 4), deadline refers to the time assigned to the broker by a user to complete a task, $\mathrm{Task_{MIPS}}$ is actual number of MIPS required by a task and $\mathrm{VM_{Available\_MIPS}}$ is total number of free MIPS available to a virtual machine. For our proposed algorithm, assigned time to broker must be greater than the ratio of actual number of MIPS required by a task and total number of free available MIPS on virtual machine.

If the datacenter has a large number of VMs, then it consumes more resources of Host. So, if datacenter having less workload, then utilize minimum resources of datacenter by reducing number of VMs in system and when workload increase then increase the number of VMs in datacenter. Dynamic VM creation algorithm executing in specific time interval. Dynamic VM creation algorithm is running for all types of VMs and manages all three types of VMs in datacenter.

Algorithm 1 Task Submission Algorithm

SortedTasklist <– null
Sort all task in increasing order of deadline and store in SortedTasklist
For each task t in SortedTasklist
    For each VM in VM list of same categories
        If task deadline > (mips required by task / free mips of vm)
            Submit task t on vm
        End If
    End for
End For

For the above algorithm, sorted tasks (increasing order of deadline) are stored in a queue. Each task from the queue is selected and if the deadline for that task is greater than the ratio of the number of MIPS required by the task and the total number of free available MIPS on the virtual machine, then that task is submitted to that virtual machine. In our proposed algorithm, this process will be repeated for each task in the queue and checked for each virtual machine in the virtual machine list.

A. ***Dynamic VM Creation***

The number of VM should be optimum so that it can maintain the quality of service parameters and consumes less energy. If number of VM is more, it consumes more power and if number of VM is less, it will violate the Service Level Agreement (SLA) and degrade the performance of the system. Hence, when more than 70% of VM is overloaded, scheduler will create a new VM. This condition is represented by equation 5.

$$Number of overloaded VM > 0.7 * total VM create a new VM \quad (5)$$

Once the task completes its execution, virtual machine will be freed, therefore some of the virtual machines need to be switched off to get optimum results. Hence

$$Number of overloaded VM < 0.3 * total VM \quad (6)$$

Therefore, a virtual machine is overloaded if it holds equations 7), (8) and (9) true:

$$TU_M > TH_M \quad (7)$$

$$TU_R > T \quad (8)$$

$$TU_B > TH_B \quad (9)$$

Here, TU_M is representing the MIPS actual utilization and TH_M is representing the threshold of MIPS. TU_R is actual RAM utilization and TH_R is threshold value of RAM utilization. TU_B is actual Bandwidth utilization and TH_B is threshold value of Bandwidth utilization. Whenever the actual utilization of MIPS, memory and bandwidth by a virtual machine is exceeding the threshold value of MIPS, memory and bandwidth of virtual machine, then virtual machine is considered overloaded.

A. ***VM Migration***

For effective resource utilization, continuous VM migration must take place. To do so, resource utilization by hosts must be estimated. So, for every host, RAM, MIPS and bandwidth utilization must be calculated based on the equation 10, 11 and 12. The RAM utilization is calculated by dividing the total RAM utilization of all virtual machine by RAM utilization of physical machine. Similar way, for each host, MIPS and Bandwidth utilization is also calculated. Average of RAM, MIPS and bandwidth utilization by a host provides the estimate of overall resource utilization by a host (equation 13).

$$TU_R = \frac{\sum_{j=0}^{m} R_{vj}}{R_{pm}} \quad (10)$$

$$TU_M = \frac{\sum_{j=0}^{m} M_{vj}}{M_{pm}} \quad (11)$$

$$TU_B = \frac{\sum_{j=0}^{m} B_{vj}}{B_{pm}} \quad (12)$$

$$TUR = [TU_R + TU_M + TU_B]/3 \quad (13)$$

Where,

TUR = Total Utilization of the Resource

TU_R = Utilization of RAM (Memory)

TU_M = Utilization of MIPS (CPU)

TU_B = Utilization of Bandwidth (Network)

For VM migration, following condition must satisfy,

$$TUR > TH \quad (14)$$

Where, TH is the threshold value of the resource utilization. If equation 14 holds true, then VM will be migrated to the balanced server, where balanced server utilization is in the range of lower and upper bound of threshold.

In the algorithm2, there is no physical machine selected at the beginning. Numbers of VMs are set as previous day total VMs. Next step is to count the total number of overloaded virtual machines in the VM list. A new VM will be created if the total number of overloaded VM is exceeding 70%. This newly created VM will increase the total number of VM by 1. Conversely, if the total number of overloaded VM falls below 30%, then least loaded VM will be terminated after migrating load from that VM and so, reducing the number of overloaded VM by 1.

Algorithm 2 Dynamic VM Creation Algorithm

Input: Initial_VM = Previous day vms total
Output: Ensure optimal number of vms in datacenter
Selected_PM ß NULL
No_of_VMs ß Initial_VM
For each VM v_i in VMList
  If (isoverloaded (v_i))
    Count++
  End if
End for
*If (No_of_VMs > count*0.7)*
  create a new VM
  Find host which has least load from running host to initiate VM
  No_of_VMs++
End if
*Else if (No_of_VMs < count*0.3)*
Migrate load from least loaded VM and terminate that VM
  No_of_VMs–
End elseif

4 PERFORMANCE ANALYSIS

In this section we have analyzed the performance of our proposed algorithms. In a highly dynamic environment like cloud it is not possible to do experiments in a repeatable and dependable fashion. That's why we have chosen the CloudSim simulation tool [42] to test our algorithms before deploying them in real cloud. CloudSim supports modeling and simulation of data centers on a single physical computing node.

The total energy consumption of the datacenter is the performance measure of greatest interest to many green cloud researchers. Energy consumed by the DBDV, Threshold based method, Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) algorithms in a datacenter consisting of 100, 200, 300, 400, 500 and 1000 cloudlets is illustrated in Figure 3, respectively. In each graph,

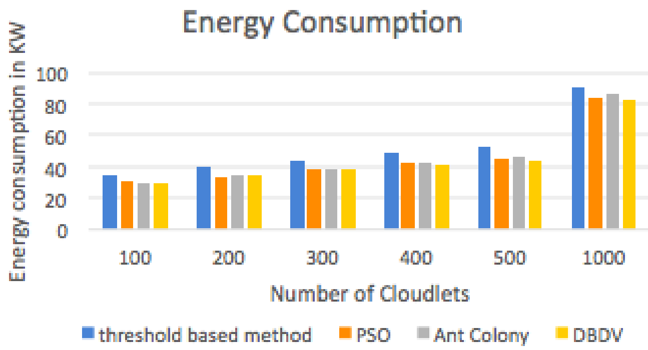

Figure 3: Energy Consumption Comparison

the X-axis represents the load of the datacenter in terms of the number of cloudlets multiplied by 100, and the Y-axis represents energy consumed in KW/hour. The total energy consumption by threshold-based method is 14-22% more than the energy consumption by the proposed DBDV method. The energy consumption by the PSO method is 1.3-4.6% more than the energy consumed by DBDV method and finally the energy consumed by the ACO algorithm is 0.7 -7% more than the energy consumed by DBDV. Overall, DBDV improved datacenter energy consumption levels to a much greater degree than threshold-based method, PSO and ACO when considering all datacenter scales and loads.

The total number of VM migration in the datacenter is another performance measure of many green cloud researchers. Number of VM migration taking place by the DBDV, Threshold based method, Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) algorithms in a datacenter consisting of 100, 200, 300, 400, 500 and 1000 cloudlets is illustrated in Figure 4, respectively. In each graph, the X-axis represents the load of the datacenter in terms of the number of cloudlets multiplied by 100, and the Y-axis represents number of VM migration. The VM migration in threshold-based method is 14-18% more than the VM migration in proposed DBDV method. The VM migration in PSO method is 7-10% more than the VM migration in DBDV method and finally the VM migration in ACO algorithm is 10.5 -13% more than the VM migration in DBDV. Overall, DBDV improved datacenter VM migration level to a much greater degree than threshold-based method, PSO and ACO when considering all datacenter scales and loads.

The completion time of a task by virtual machine is one of the key performance measures for researchers. Completion time of a task by DBDV, Threshold based method, PSO and ACO in a datacenter consisting of 100, 200, 300, 400, 500 and 1000 cloudlets is illustrated in Figure 5. In the graph, X - axis represents the load of the datacenter in terms of number of cloudlets which is multiple of 100 and the Y – axis represents completion time for a task. The completion time by threshold-based method is 9-17% more than the completion time by proposed DBDV method. The completion time by PSO method is 1.7-6% more than the completion time by DBDV method and the completion time by ACO algorithm is 4.6 -10% more than the completion time by DBDV. The completion time graph shows that DBDV takes less time to complete the task than threshold-based method, PSO, ACO.

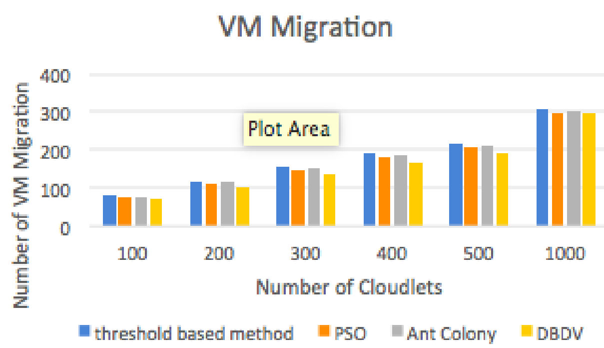

Figure 4: VM Migration Comparison

5 CONCLUSION AND FUTURE WORK

This paper intended to provision green cloud computing concepts by restricting the amount of energy consumption in cloud datacenters. The suggested algorithm, DBDV, was motivated by locust phase change behavior. Similarly, the DBDV algorithm divided two segments: first segment is task submission on VM and second segment is VM creation and migration on host. In the task submission segment, task submitted on VMs and ensures that each task completed within the deadline while in the VM creation and creation and migration segment, VMs are created and migrated on host to optimize energy consumption. The experimental results show that DBDV have improved resource utilization levels to a much greater degree than two other benchmarks by reducing the number of VM migration involved. In terms of energy consumption levels, on balance DBDV improved energy consumption levels more than the benchmarks with respect to all datacenter loads and scales, though ThrMu applied at large datacenter scales was found to be an exception. In future work we plan to evaluate DBDV's performance by considering the network traffic of datacenter components. DBDV does not consider server clusters. It is our intention to consider server clusters like Google clusters in future work. We plan to use LLS as a LIGHT server with VMs requiring the lowest amount of migration time within the cluster and to determine how this affects DBDV performance. Additionally, we plan to experiment with

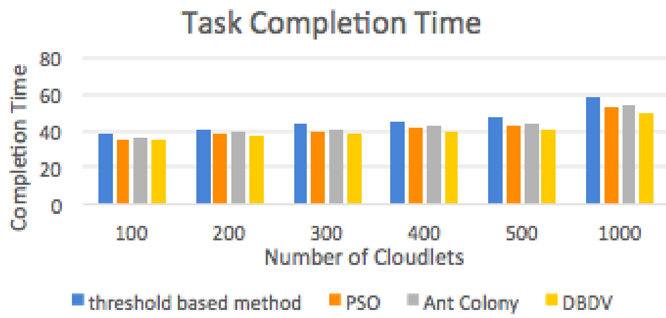

Figure 5: Completion Comparison

DVFS servers rather than regular servers to save even more energy. We also intend to measure the effects of burst workload arrivals.

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